EXP. NO. 07

DECISION TREE CLASSIFIER USING ID3

AIM

Implement a Decision Tree classifier using the ID3 algorithm to segment customers based on their purchasing behavior using the Online Retail dataset. Analyze the tree structure and discuss the feature importance.

Tasks

- Load and preprocess the Online Retail dataset.
- Implement a Decision Tree using the ID3 algorithm.
- Create customer-level purchasing features using RFM (Recency, Frequency, and Spent).
- Segment customers into Low Spent and High Spent categories.
- Visualize the decision tree and analyze feature importance.
- Discuss how the tree structure helps in understanding customer behavior.

THEORY

Decision Tree is a supervised machine learning algorithm used for classification and regression. It represents decisions in the form of a tree structure consisting of root nodes, internal nodes, branches, and leaf nodes.

The **ID3 (Iterative Dichotomiser 3)** algorithm constructs a decision tree using **Entropy** and **Information Gain**. The feature having the highest information gain is selected for splitting the data.

Entropy

Entropy measures the impurity or uncertainty in a dataset. It is calculated as:

$$Entropy(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

where p_i is the proportion of samples belonging to class i.

For a binary classification problem:

$$Entropy(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

Information Gain

Information Gain measures the reduction in entropy after splitting the dataset based on a feature.

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

The feature with the highest Information Gain is selected as the decision node.

In this experiment, the Decision Tree is implemented using the `entropy` criterion, which follows the information-gain principle of the ID3 algorithm.

Customer Segmentation using RFM

The Online Retail dataset contains transaction-level information. Customer-level purchasing behavior is obtained using three features:

- **Recency** – Number of days since the customer's most recent purchase.
- **Frequency** – Number of unique invoices or orders made by the customer.
- **Spent** – Total amount spent by the customer.

The spending for each transaction is calculated as:

$$Spent = Quantity \times UnitPrice$$

The median spending is used to create two customer segments:

- **High Spent** – Customer spending is greater than or equal to the median spending.
- **Low Spent** – Customer spending is less than the median spending.

The Decision Tree learns relationships between customer purchasing behavior and the customer segment.

ALGORITHM

Step 1: Load the Online Retail dataset from the CSV file using Latin-1 encoding.

Step 2: Remove records with missing `CustomerID` values.

Step 3: Remove transactions with non-positive `Quantity` values.

Step 4: Convert the `InvoiceDate` column into date-time format.

Step 5: Calculate the total spending for each transaction using:

$$Spent = Quantity \times UnitPrice$$

Step 6: Determine the reference date using the latest `InvoiceDate` in the dataset.

Step 7: Group the transactions according to `CustomerID` to create customer-level RFM features.

Step 8: Calculate Recency, Frequency, and Spent for each customer.

Step 9: Calculate the median value of customer spending.

Step 10: Create the customer segment:

- High Spent if spending is greater than or equal to the median.
- Low Spent otherwise.

Step 11: Separate the input features and target variable.

Step 12: Split the dataset into training and testing sets.

Step 13: Train a Decision Tree classifier using the Entropy criterion.

Step 14: Predict the customer segments for the test data.

Step 15: Evaluate the classifier using accuracy, precision, recall, and F1-score.

Step 16: Visualize the Decision Tree.

Step 17: Calculate and analyze the feature importance of the purchasing-behavior features.

EXPECTED OUTCOME

The Decision Tree classifier should classify customers into **Low Spent** and **High Spent** categories based on their purchasing behavior. The visualized tree will show the decision rules, while feature importance will indicate which purchasing-behavior features contribute most to the classification.